

The Gauge Algebra of Chirality

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We classify adjoint complements of ordered maximal two-simple-factor subalgebras of compact simple Lie algebras by complex type and cubic trace.

The domain and the two tests

Let $\mathfrak{g}$ be compact and simple, and let $\mathfrak{h} = \mathfrak{k} \oplus \mathfrak{l} \subsetneq \mathfrak{g}$ have *exactly two nonabelian simple ideals*. Require $\mathfrak{h}$ to be maximal among all proper Lie subalgebras, not merely such products. Both orderings are tested, with $\mathfrak{k}$ the tested factor. The invariant orthogonal complement $\mathfrak{q} = \mathfrak{h}^\perp$ is an $\mathfrak{h}$ -module.

Decompose $\mathfrak{q}_\mathbb{C}$ into *joint* $\mathfrak{h}$ -irreducibles $R \boxtimes S$. Put all summands with $R \simeq R^*$ in q_0 . The rest occur in dual pairs with equal multiplicities:

$$\mathfrak{q}_\mathbb{C} = q_0 \oplus \bigoplus_{\alpha} m_{\alpha} (U_{\alpha} \oplus U_{\alpha}^*).$$

Choose one member of each such pair, retaining its *full multiplicity*, and call their sum W . Every choice is checked. Thus W is $\mathfrak{h}$ -invariant: one may not independently reorient the $\mathfrak{k}$ -copies inside an $\mathfrak{l}$ -multiplet. The tests are

$$(A) \quad W \neq 0, \quad (B) \quad d_{\mathfrak{t}}(W)(X, Y, Z) = \frac{1}{2} \text{Tr}_W(T_X\{T_Y, T_Z\}) = 0 \quad (X, Y, Z \in \mathfrak{k}),$$

where $T_X = -i\rho_W(X)$ is Hermitian. Test (B) is the vanishing of the local cubic anomaly tensor on $\mathfrak{k}$.

Exclusion at every classical rank

The classification [1,2] gives parameterized families of unbounded rank. Unitary block stabilizers have a central $\mathfrak{u}(1)$, outside this domain. Orthogonal and symplectic blocks have tensor-product defining complements with self-dual constituents. The tensor-embedding families are

$$\begin{aligned} \mathfrak{su}(p) \oplus \mathfrak{su}(q) &\subset \mathfrak{su}(pq), & \mathfrak{so}(p) \oplus \mathfrak{so}(q) &\subset \mathfrak{so}(pq), \\ \mathfrak{sp}(p) \oplus \mathfrak{sp}(q) &\subset \mathfrak{so}(4pq), & \mathfrak{so}(p) \oplus \mathfrak{sp}(q) &\subset \mathfrak{sp}(pq). \end{aligned}$$

Here $\mathfrak{sp}(p)$ has defining dimension $2p$. Retain only maximal inclusions satisfying the simplicity requirements, including low-rank isomorphisms. The unitary tensor complement is $\text{ad } \mathfrak{su}(p) \boxtimes \text{ad } \mathfrak{su}(q)$. In the other cases use

$$\Lambda^2(U \otimes Z) = (\Lambda^2 U \otimes \text{Sym}^2 Z) \oplus (\text{Sym}^2 U \otimes \Lambda^2 Z)$$

and the analogous formula for $\text{Sym}^2(U \otimes Z)$, with both symmetric squares together and both exterior squares together. After subtracting the two adjoints, the factorwise constituents are self-dual: adjoints and traceless symmetric or exterior squares of defining modules. Hence every classical row has $W = 0$, at arbitrary rank.

Exceptional elimination

The exceptional branchings [3] leave the five unordered embeddings on page 2. In particular, for $E_6 \supset A_2 + G_2$, with factors in that order,

$$78 = (8, 1) \oplus (1, 14) \oplus (8, 7).$$

Both factors see only self-dual constituents. The complex $(3, 7)$ belongs instead to $27 = (\bar{6}, 1) \oplus (3, 7)$. The $A_5 + A_1$ complement $(20, 2)$ is also factorwise self-dual. Neither embedding passes (A).

The five complex rows and the survivor

Dynkin types label the factors below. Entries are cubic coefficients in the convention $A(\mathbf{n}) = 1$ for $\mathfrak{su}(n)$; dualizing a whole half reverses their signs. The reverse column also reapplies test (A).

g	($\mathfrak{f}, \bar{\mathfrak{f}}$)	Half W	$\mathfrak{f}$ test	Reverse test
F_4	(A_2, A_2)	$(\mathbf{6}, \bar{\mathbf{3}})$	21	−6
E_7	(A_5, A_2)	$(\mathbf{15}, \bar{\mathbf{3}})$	6	−15
E_8	(A_4, A_4)	$W_{a,b}$	$5a + 10b$	$-10a + 5b$
E_8	(A_2, A_1)	$(\mathbf{10}, \mathbf{5})$	135	(A) fails
E_8	(E_6, A_2)	$(\mathbf{27}, \mathbf{3})$	0	27

For the $A_4 + A_4$ row, put $U = (\mathbf{10}, \bar{\mathbf{5}})$, $Z = (\mathbf{5}, \mathbf{10})$, and $W_{a,b} = U^{(a)} \oplus Z^{(b)}$, where $a, b = \pm 1$ and $R^{(+1)} = R$, $R^{(-1)} = R^*$. Both coefficient sets are $\{\pm 5, \pm 15\}$. For the special $A_2 + A_1 \subset E_8$ embedding, the remaining complement is $q_0 = (\mathbf{8}, \mathbf{7}) \oplus (\mathbf{27}, \mathbf{3})$, with self-dual A_2 -constituents; in reverse order every A_1 -representation is self-dual.

The calculations use $A(R \boxtimes S) = \dim(S)A(R)$, additivity and duality, with $A_{A_2}(\mathbf{6}) = 7$, $A_{A_5}(\mathbf{15}) = 2$, and $A_{A_2}(\mathbf{10}) = 27$. The E_6 coefficient vanishes because its Lie algebra has no invariant symmetric cubic. This is distinct from the cubic on $\mathbf{27}^{\otimes 3}$, which has representation rather than Lie-algebra indices.

Theorem 1 (Ordered maximal-product classification). *In the stated domain and with the stated half-selection rule, (A) and (B) select $\mathfrak{e}_8 \supset \mathfrak{e}_6 \oplus \mathfrak{su}(3)$, with tested factor $\mathfrak{f} = \mathfrak{e}_6$, uniquely up to isomorphism of ordered Lie-algebra pairs. The half is $W = \mathbf{27} \boxtimes \mathbf{3}$, up to duality.*

Restriction and net chiral content

The adjoint branching is

$$\mathbf{248} = (\mathbf{78}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{8}) \oplus (\mathbf{27}, \mathbf{3}) \oplus (\bar{\mathbf{27}}, \bar{\mathbf{3}}).$$

Consequently $W|_{E_6} = 3 \mathbf{27}$. At group level use

$$G_{\text{SM}} = [SU(3) \times SU(2) \times U(1)]/\mathbb{Z}_6 \subset SU(5) \subset \text{Spin}(10) \subset E_6^{\text{sc}}.$$

Restriction gives

$$\mathbf{27}|_{\text{Spin}(10)} = \mathbf{16} \oplus \mathbf{10} \oplus \mathbf{1}, \quad \mathbf{27}|_{SU(5)} = \underbrace{\mathbf{10} \oplus \bar{\mathbf{5}}}_{\mathcal{F}} \oplus (\mathbf{5} \oplus \bar{\mathbf{5}}) \oplus \mathbf{2} \mathbf{1}.$$

Here $\mathcal{F}|_{G_{\text{SM}}}$ is one minimal family:

$$(\mathbf{3}, \mathbf{2})_{1/6} \oplus (\bar{\mathbf{3}}, \mathbf{1})_{-2/3} \oplus (\bar{\mathbf{3}}, \mathbf{1})_{1/3} \oplus (\mathbf{1}, \mathbf{2})_{-1/2} \oplus (\mathbf{1}, \mathbf{1})_1.$$

The labels use the all-left-handed convention; the singlet in the $\mathbf{16}$ is the additional ν^c , not part of this minimal family.

Corollary 2 (Containment). *The selected half restricts to three minimal Standard Model families, three vectorlike $\mathbf{5} \oplus \bar{\mathbf{5}}$ pairs, and six singlets. Its net chiral class is $3\chi(\mathcal{F}|_{G_{\text{SM}}})$; the dual half has the opposite class.*

Here χ records representation-minus-dual multiplicities and annihilates self-dual summands.

References. [1] E. B. Dynkin, *Semisimple subalgebras of semisimple Lie algebras*, Mat. Sb. **30** (1952), 349–462; *Maximal subgroups of the classical groups*, Tr. Mosk. Mat. Obs. **1** (1952), 39–166. [2] F. Antoneli, M. Forger and P. Gaviria, *Maximal Subgroups of Compact Lie Groups*, arXiv:math/0605784, Tables 2–4. [3] R. Slansky, *Group theory for unified model building*, Phys. Rep. **79** (1981), 1–128, doi:10.1016/0370-1573(81)90092-2, Tables 15 and 58.